

JSS 3 MATHEMATICS – FIRST TERM

TOPIC 4: MULTIPLICATION AND DIVISION OF NUMBERS IN BASE 2

1. Binary Multiplication

Multiply as in base 10: $0 \times 0 = 0$, $0 \times 1 = 0$, $1 \times 0 = 0$, $1 \times 1 = 1$. Shift left one place for each next digit of the multiplier, then add the partial products in binary.

2. Example

$11 \blacksquare \times 10 \blacksquare$: $11 \times 0 = 00$; $11 \times 1 = 11$ (shifted one place) $\rightarrow 11 \blacksquare + 110 = 110 \blacksquare$. Check: $3 \times 2 = 6$, and $110 \blacksquare = 6$.

3. Binary Division

Divide as in long division. Binary division is often simpler because the digit is only 0 or 1. Subtract (in binary) when the divisor fits; bring down the next digit and continue.

4. Summary

Multiplication and division in base 2 use the same structures as base 10, with binary addition/subtraction for the partial steps. Checking in base 10 builds confidence.

End of Topic 4